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A hybrid method for data evaluation in 1-D Raman spectroscopy

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Abstract

In turbulent combustion research, 1-D Raman/Rayleigh spectroscopy is applied to understand turbulence-chemistry interaction by simultaneously measuring the major species concentrations and temperature. Two different methods have been used in the past to evaluate the data: spectral fitting, based on libraries of theoretical spectra; or matrix inversion of signals that are not spectrally resolved, based on extensive calibration of temperature-dependent system response. A hybrid method for Raman/Rayleigh data analysis is presented that combines the strengths of both previous methods. This hybrid approach is demonstrated using measurements in methane/air flat flames and a laminar hydrogen jet diffusion flame. Precision and accuracy of results from the various methods are compared, and benefits of the hybrid method are discussed. The hybrid approach described here may be adapted to any Raman/Rayleigh detection setup that is optically well characterized.
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1. Introduction

Multiscalar measurements in turbulent flames are an important key to a detailed understanding of turbulence–chemistry interaction. Raman/Rayleigh spectroscopy is a valuable diagnostic tool for this purpose [1–4]. Spontaneous Raman scattering does not require tunable excitation radiation, and the spectrum of Raman scattered light contains characteristic signatures of different molecules in the probe volume. The scattering process is not disturbed by inter-molecular collisions, so signal evaluation is independent of bath conditions. These advantages are offset by small scattering cross sections, such that single-shot Raman scattering in turbulent hydrocarbon flames allows only the detection of main species (e.g., H_2, O_2, N_2, CO, CO_2, H_2O, CH_4), based on rovibrational Raman transitions of the Stokes side.

Spontaneous Raman scattering, applied by itself or in combination with Rayleigh scattering of the same laser source, has been used to study a broad range of turbulent flames [4,5]. Much of this work has been motivated by the need for detailed data to validate combustion models. In recent years, this has included measurements of scalar gradients and scalar dissipation in atmospheric pressure CH_4/air flames, including nonpremixed,

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partially premixed, stratified, and premixed cases [5–7]. Gradient measurements demand higher spatial resolution and precision than is needed to measure statistics of the scalars themselves [8]. This has prompted development of specialized imaging spectrographs using high efficiency transmission gratings and low *f*-number optics [9]. Commensurate with the improved resolution and detection efficiency of these systems is the need to optimize data acquisition and processing methods to achieve the best possible accuracy and precision.

The processing of Raman scattering signals into concentrations must account for the temperature dependence of Raman spectra and the integrated scattering cross sections of each measured species. Raman spectra of certain species overlap. For example, there is crosstalk between CO_2 and O_2, between CO and N_2, and from H_2 rotational lines onto all the other species to varying degrees [4,10–14]. In addition, there can be interferences from laser induced fluorescence and from broadband flame luminosity [4,12,13].

Two different approaches to Raman data processing are described in the turbulent combustion literature. In the matrix inversion method (MI) [4,10–12] Raman signals, fluorescence interferences, and background radiation are integrated across fixed spectral regions (Raman channels) to form a signal vector S. Spectral information on broadening and shifting of Raman bands with increasing temperature is lost, as are spectral details of interferences. When a CCD is used for detection, Raman channels are defined by on-chip binning, whereas in earlier approaches photomultiplier tubes were placed at fixed spectral positions at the exit plane of a large spectrometer. The signal vector, S, is related to the corresponding vector of unknown species number densities, N, through the matrix equation $S = P(T)N$, where the matrix elements depend on temperature and the specific experimental setup. Diagonal elements $p_{ii}(T)$ are the Raman responses of the various species, whereas off-diagonal elements $p_{ij}(T)$ represent crosstalk between Raman channels and from the fluorescence interferences and background luminosity channels. The temperature dependence of each matrix element is represented as a polynomial and based on calibration measurements in cold gases, electrically heated gases, and well characterized laminar flames. Number densities in turbulent flames are determined by solving the inverse equation for the vector N. The local instantaneous temperature from simultaneously measured Rayleigh scattering is used in evaluating the matrix elements. Because the effective Rayleigh cross-section depends on the gas composition, an iterative solution is applied.

The second approach for processing Raman data is the spectral fitting method (SF) [15]. Based on Placzek's theory of polarizability, individual rovibrational Raman transitions are calculated for each species. Raman bands broaden with increasing temperature due to the population of higher quantum states. Each Raman transition is convolved with an experimentally determined apparatus function of the detection unit. This apparatus function dominates all other linewidth effects. Finally, all convolved Raman transitions are superposed, resulting in rovibrational Raman bands. A spectral library is composed of these temperature-dependent Raman bands.

The spectral library for each molecule is then calibrated to an experimental spectrum measured in a gas sample with known mole fraction and temperature. This accounts for specific experimental conditions (detection and quantum efficiency, etc.). Subsequently, species concentrations in turbulent flames can be computed directly by fitting measured spectra with the calibrated libraries using an advanced routine [15]. Crosstalk is taken into account inherently, and the background is accounted for by fitting it simultaneously with the Raman spectra, using piecewise cubic splines or linear interpolation.

Each method for processing Raman data has advantages and disadvantages. The matrix inversion method allows for lower readout noise due to on-chip binning, and processing is fast. However, extensive calibration is needed, and uncertainties in calibrations of reactive species and crosstalk terms at flame temperatures can be large. Furthermore, it is not practical to account for the effects of beam steering in turbulent flames, which shifts the entire Raman spectrum relative to the fixed on-chip binning regions. On the other hand, the spectral fitting method is based on quantum mechanical theory. This reduces the number of calibrations to only one per species and reduces uncertainty within temperature ranges that are difficult to calibrate. For luminous flames the background radiation can be corrected more rigorously in the spectral fitting process, and beam steering can be automatically handled. However, the preserved spectral information comes with the price of higher readout noise, slower data acquisition rates, and significant effort fitting the Raman spectrum of each single-shot measurement.

This paper presents a hybrid approach for processing Raman data, combining the advantages of both methods. In this hybrid approach the temperature dependence of each matrix element for Raman spectral response and crosstalk is determined by integrating the calculated spectral libraries over regions corresponding to on-chip binning. In the following sections the experimental setup and hybrid method are described, calculated and measured response curves are compared, and the different post-processing methods are evaluated using results from premixed methane/air flat flames and a laminar hydrogen jet diffusion flame.

2. Experimental setup and hybrid method

The experimental setup for line-imaged multiscalar measurements at Sandia combines commercial camera lenses with a high efficiency transmission grating to yield significant improvements in spatial resolution and optical throughput over the previously used Czerny-Turner spectrograph [9]. The combined laser energy was ~1.8 J/pulse at 532 nm, with a beam width of roughly 0.15 mm (FWHM) and overall pulse duration of ~0.5 μs. The imaged region was 6 mm in length along the waist of the laser beam. On-chip binning was applied in the Raman measurements to give a sample spacing of 103 μm along the laser axis. Rayleigh images were acquired with 20 μm pixel resolution (binning 2 × 2). A high-transmission linear polarizer was placed in front of the spectrograph to reduce the relative importance of fluorescence interference and chemiluminescence. Two sets of data were acquired, one with no binning in the spectral direction on the Raman camera and the other with wide regions of on-chip spectral binning corresponding to Raman bands for different molecules.

An intrinsic feature of this type of spectrometer is that the image of a straight entrance slit or laser beam is optically distorted or bowed [16]. This is illustrated in Fig. 1, which shows the curved image of N_2 Raman scattering from the laser beam in room temperature air. With the present system the location of the peak in the N_2 spectrum shifts by as much as +10 pixels, comparing the lower edge of the imaged region to the optical center. A shift of +10 pixels is equivalent to a shift of 1.2 nm toward the red, based on dispersion of 5.95 nm/mm at that wavelength. The problem presented by this bowing effect is similar to that caused by beam steering in turbulent flames. In both cases the imaged Raman scattering spectrum shifts relative to fixed binning regions on the CCD. Measurements of N_2 and H_2O are relatively insensitive to these effects because binning regions are wide enough to include nearly the entire Raman spectrum at temperatures of interest. However, other species results can be quite sensitive, as will be shown. It would be impractical to account for bowing or beam steering using calibration-based polynomials because extensive calibrations would have to be repeated for many different positions of the laser beam. However, both effects can be addressed in a straightforward manner by the hybrid method.

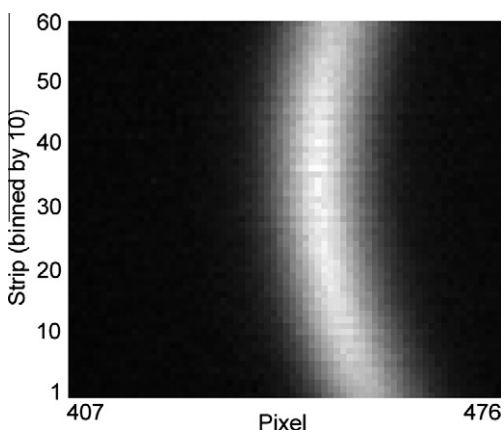

Fig. 1. Illustration of the bowing effect in the transmission spectrometer showing the N_2 Raman band measured in air at 293 K. Vertical direction displays the spatial coordinate (60 strips corresponding to 6 mm along the laser beam) and horizontal direction the wavelength (pixel) coordinate. Binning by 10 is applied in spatial direction but full resolution for the spectral dimension. The width of the image (pixel range 407–476) corresponds to the on-chip binning region for N_2.

Figure 2 shows relative Raman scattering spectral intensities calculated for CO_2, O_2, CO, N_2, H_2O and H_2 using an extended version of RAMSES presented in [15]. CH_4 is excluded from these calculations because reliable quantum mechanical models for CH_4 covering the whole temperature range are not available. For each chemical species the perpendicular polarization direction of the Raman bands are calculated for temperatures ranging from 250 to 2500 K. Calculated spectral libraries account for the measured nonlinear dispersion of the spectrometer. The Rayleigh scattering image of the laser beam, appropriately scaled, was used as the convolution apparatus function. The temperature dependence of the spectrum of each molecule and the overlap of spectra from different molecules are clearly illustrated in Fig. 2. Dashed vertical lines indicate the pixel boundaries for on-chip binning regions (Raman channels) used in the present experiments.

To demonstrate that the quantum mechanical model reproduces experimental spectra reliably, Fig. 3 shows measured and calculated N_2 and H_2O Raman bands at 2000 K. Agreement for N_2 is very good. Deviation close to the noise level for H_2O is observed.

The key to the hybrid method is to use these calculated spectral libraries rather than extensive calibrations to determine the temperature dependence of matrix elements in the MI method. This is done by integrating each calculated spectrum over the pixel regions of on-chip binning. For example, the term for crosstalk from CO_2 onto the O_2 channel is determined by integrating the calculated CO_2 Raman spectrum at each library temperature over the pixel range of the O_2 channel. Furthermore, to allow for automatic compensation for the optical bowing effect and for beam steering in turbulent flames, the integration is performed multiple times with the pixel boundaries being shifted progressively relative to the calculated spectra. Results are stored in tabular form

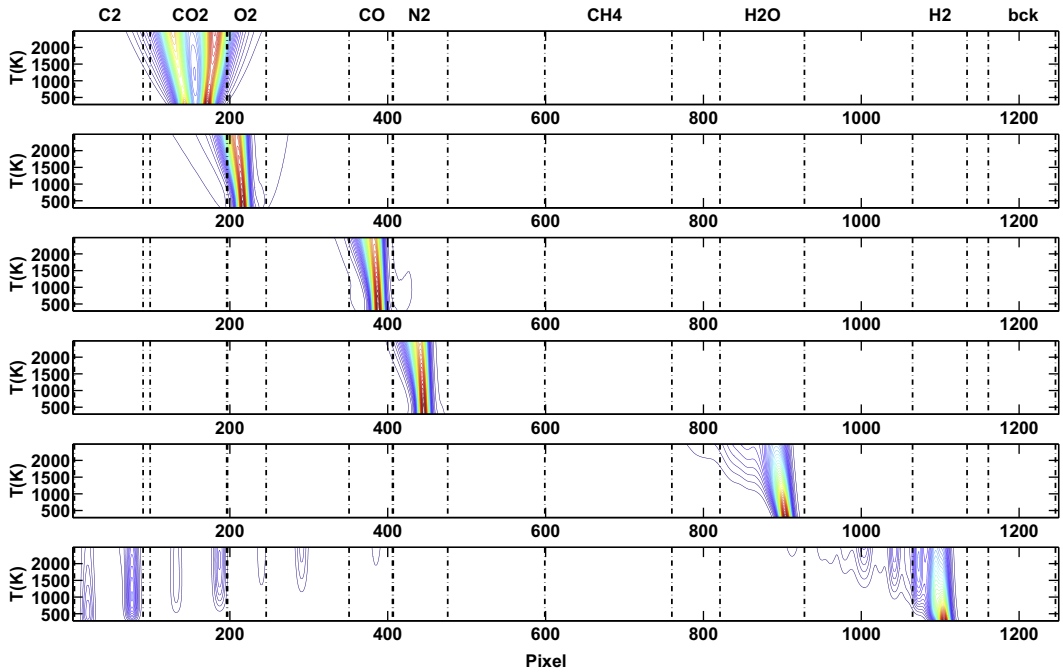

Fig. 2. Calculated temperature dependence of relative Raman scattering spectral intensities from CO_2, O_2, CO, N_2, H_2O, and H_2. Dashed vertical lines indicate the pixel boundaries for on-chip binning regions (Raman channels) used in the present experiments.

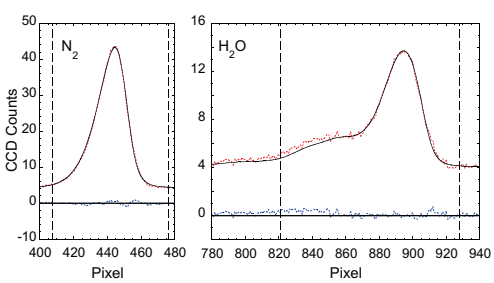

Fig. 3. Rovibrational Raman spectra for N_2 and H_2O at 2000 K in post-flame region of a lean CH_4/air flat flame. The measured spectra (dashed) are averaged using 100 shots. The calculated spectra (solid) are fitted simultaneously with a background to the experimental spectra, the residuum after subtraction is shown. Dashed vertical lines mark the binning region.

and normalized by the integrated intensity at 290 K and zero pixel shift. The calibration factor for each matrix element is determined from one or a few measurements at well define conditions of temperature and composition. During the iterative solution of the inverse matrix problem, the table of integrated relative intensities is interpolated in two dimensions, based on the local temperature and pixel shift. For the present implementation, table entries are spaced by 20 K in temperature and by one in pixel shift. Pixel shift due to the bowing effect is represented by a second order fit, and pixel shift of the overall beam due to alignment drift or beam steering is determined from the Rayleigh scattering image of the laser beam profile on each shot. Matrix elements that are not calculated theoretically, such as CH_4 and it's crosstalk onto O_2, are currently evaluated using calibrated polynomials from the original MI method.

This procedure can be extended to include interferences from C_2 LIF. It can also be extended to include hydrocarbons by generating experimentally based spectral libraries and integrating those over the pixel ranges of the binned Raman channels. Work is in progress to treat CH_4 and dimethyl ether (DME) in this way.

3. Results and discussion

For the main species appearing in hydrocarbon/air flames, CO_2, O_2, CO, N_2, H_2O and H_2, Fig. 4a shows a comparison of temperature-dependent response curves obtained from calibration measurements and from integration of calculated Raman spectra as described in the previous section. The calibration-based curves are also normalized by their values at 290 K. Note that the scales of the individual response curves vary significantly among species. Whereas CO_2 shows strong temperature dependence, the response for

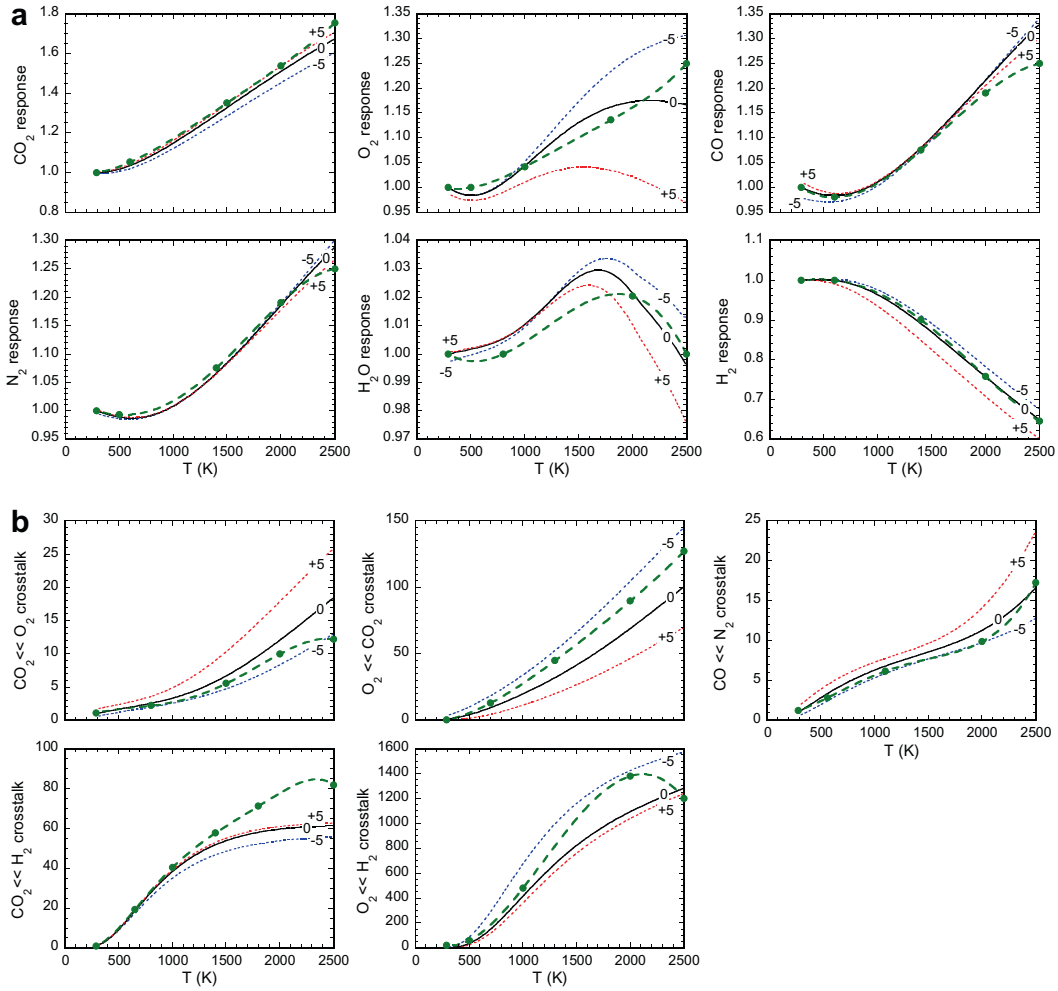

Fig. 4. Comparison of: (a) temperature-dependent Raman response curves for CO_2, O_2, CO, N_2, H_2O, and H_2, and (b) Raman crosstalk terms based on calibration polynomials (symbols show the tabulated fitting points) and integration of the calculated Raman spectra. The calculated curves are labeled 0, +5, and −5, corresponding to pixel shifts of the integration range.

H_2O barely changes with temperature and the observed discrepancy is not significant. The sensitivity of the response curves to spectral shift of the Raman channels (binning region) is illustrated by including curves for shifts of ±5 pixels, which is representative of the shift caused by the bowing effect or strong beam steering in turbulent flames.

The calculated response curves of CO_2, CO, N_2, H_2O, and H_2 are considered as the benchmark for the measured ones. Agreement between measured and calculated curves is very good for most of the temperature range. Trajectories of the response curves at rising temperatures are determined by competition between the increasing number of scattered photons and loss of signal as the spectrum of interest extends beyond the boundaries of the corresponding Raman channel due to the larger fractional population of the higher energy levels. Inspecting CO_2 intensity in Fig. 2, for temperatures exceeding approximately 800 K the Raman channel does not capture the whole Fermi-coupled $\nu 1 + 2\nu 2$ Raman bands but is still increasing because most of the transitions are within the binned region. In contrast, H_2 shows a decreasing response curve. Due to its large rotational constant the spacing between rovibrational Raman lines is much wider than for the other molecules. Consequently, due to the shift of the fractional population at higher temperatures, a smaller fraction of the overall increasing number of scattered photons is captured on the pre-defined H_2 Raman channel. For N_2 an excellent match between experimental and calculated response curves is observed, except at

very high temperatures. Similar behavior is observed for CO. For O_2 there is a mismatch between shapes of experimental and calculated response curves at high temperature. Uncertainty in calibration increases at flame temperatures, especially for reactive species, because concentrations are low and luminosity can contribute significantly to the measured signal.

The results for the more important off-diagonal elements accounting for Raman crosstalk are presented in Fig. 4b. For example, the contribution to the CO_2 Raman channel coming from O_2 is labeled as $CO_2 << O_2$. Note that the scale on each plot indicates relative change over the temperature range rather than absolute magnitude because the curves are normalized to unity at 290 K. The most important crosstalk term in hydrocarbon flames is $O_2 << CO_2$ because of significant spectral overlap (Fig. 2) and low O_2 concentration at high temperature. Although the overall trends between measured and calculated response curves coincide, a high sensitivity to pixel shift is observed. The response curve for O_2 in Fig. 4a is also sensitive to pixel shift. Sensitivity of these terms demonstrates the need to account for optical bowing. Equally important is an accurate calibration of wavelength vs. camera pixel, which was accomplished here using a target array of 50 micron holes back-illuminated by a Ne lamp. Whereas the crosstalk term $CO << N_2$ can be accurately calibrated using lean H_2/air flames, uncertainty in the experimentally derived curve for the other crosstalk terms can be large. This gives significant advantage to the hybrid approach.

Figure 5 illustrates the effect of bowing on Raman crosstalk by comparing spatial profiles of the mole fraction of O_2 measured in products over a fuel-rich premixed laminar flat flame, where the actual mole fraction is practically zero.

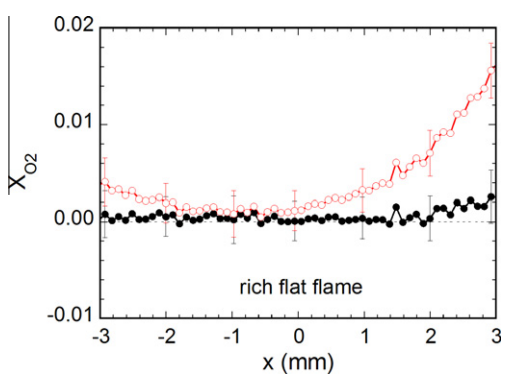

Fig. 5. Illustration of the effect of optical bowing on crosstalk from CO_2 onto the O_2 Raman channel in flat flame products at $\phi = 1.16$. Error bars indicate standard deviation in the polynomial MI method (open symbols) and the hybrid method (closed symbols).

For these spatially homogeneous conditions, the conventional MI method yields an inhomogeneous O_2 profile that is accurate only near the bowing center (−1 mm), where calibrations are tuned. Homogeneity of the results is significantly improved by application of the hybrid method to the same raw data file. Standard deviations, indicated by error bars in Fig. 5, are the same for the two methods.

Scatter plots of single-shot results for T, O_2, and CO_2 obtained using four different processing approaches are compared in Fig. 6. Measurements are from products above a premixed laminar CH_4/air flat flame operated at five different equivalence ratios from lean to rich. Data were acquired sequentially in two sets, one with no binning in the spectral direction and the other with on-chip binning as illustrated in Fig. 2. Only three equivalence ratios are shown for the cases with no binning to provide better separation. The four data evaluation methods were: MI using calibrated polynomials, hybrid MI, software binning of the spectrally resolved data followed by hybrid MI, and spectral fitting. Several observations can be made. First, with the present detection system, on-chip binning yields significant better precision at flame conditions. This is because the binned signals are limited primarily by photon shot noise and the relative contribution of camera readout noise is suppressed. Second, the effects of bowing are apparent in the polynomial MI scatter data, which show extended tails resulting from CO_2 crosstalk onto the O_2 channel. This crosstalk correlates directly with error in measured equivalence ratio. Temperature is also affected through a change in the apparent Rayleigh cross section. Specifically, bowing at the edges of the image moves CO_2 signal onto the O_2 channel, reducing the calculated Rayleigh cross section, so the same Rayleigh signal yields higher apparent total number density and lower temperature. These effects are corrected in the hybrid MI processing. Negative concentration results are prevented in the SF. Other differences between software binned MI and SF are small and probably caused by distinct calibration methods and background treatment.

Finally, the four data evaluation schemes are compared using measurements from a laminar H_2 jet flame. Figure 7 shows radial profiles of mean and rms mole fractions of N_2, O_2, and H_2O taken at an axial height of 30 mm. Profiles were acquired using 3 mm steps of the 6 mm probe, such that each radial location was measured by two different strips on the Raman CCD. Processed results were then averaged. The bowing effect causes discontinuities in mean and rms O_2 profiles from the polynomial MI method, and the software binned results for O_2 show a small offset due to an uncorrected drift in the readout offset of the CCD array, which is negligi-

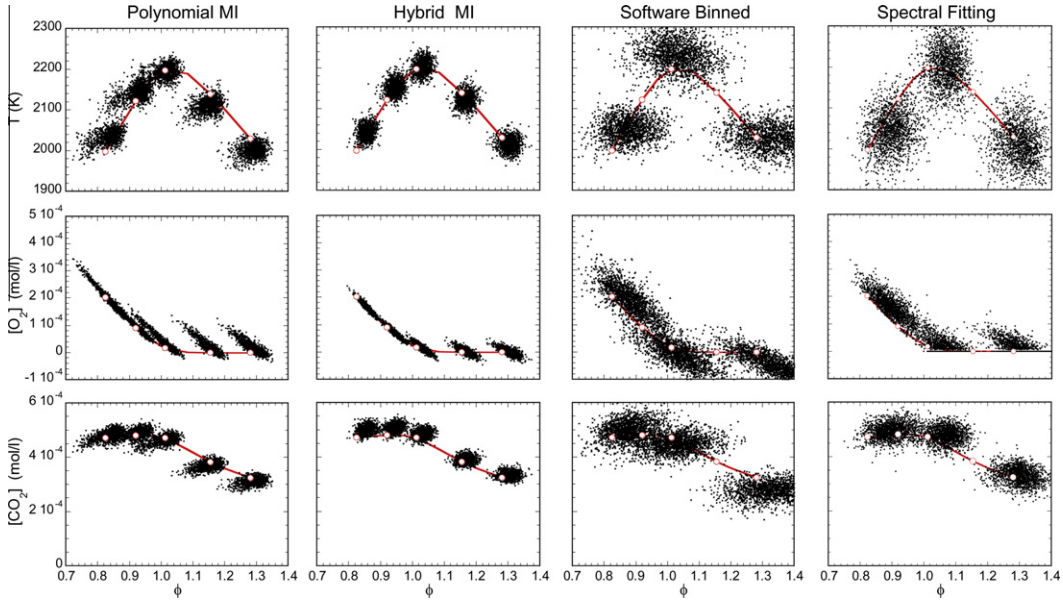

Fig. 6. Scatter plots of temperature and the concentrations of O_2 and CO_2 measured above a series of premixed CH_4/air flat flames and processed using the indicated methods: matrix inversion with polynomials, the hybrid method, the hybrid method applied to software-binned Raman data, and spectral fitting. Curves through open circles show equilibrium values at 35 K below the adiabatic equilibrium temperature and are taken from laminar flamelet calculations.

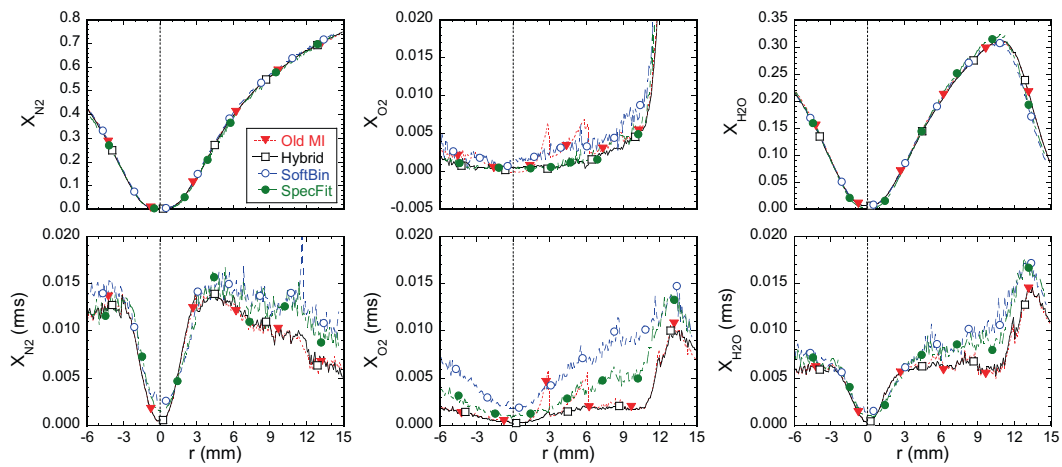

Fig. 7. Radial profiles of Favre mean and rms species mole fractions based on the polynomial matrix inversion method (Old MI), bin-on-chip hybrid method, software binned hybrid method, and spectral fitting method of data analysis.

ble for the hardware binned data. Otherwise, mean profiles are in good agreement. Greater differences among processing methods are seen in the rms profiles, where on-chip binning shows advantage across parts of the flame. For the same spectrally resolved data, the spectral fitting method yields lower rms values, especially for O_2, than software binning combined with the hybrid MI method. This shows that the fitting process can serve as a noise filter and, at least, does not add noise to the results. It is also clear that software binning can be applied to achieve rapid initial processing of spectrally resolved data.

4. Conclusions

A method of analyzing 1-D Raman/Rayleigh scattering data from flames has been developed, which combines advantages of two methods used

by the contributing institutions: (1) fitting of spectrally-resolved, single-shot data with libraries of theoretically calculated Raman scattering spectra; and (2) iterative solution of a matrix equation for integrated Raman signals (binned on the CCD detector), using polynomials to represent the calibrated temperature dependence of matrix terms. In the resulting method the temperature dependence of matrix terms was determined by integrating calculated Raman spectral libraries over intervals corresponding to regions of on-chip binning. This hybrid matrix inversion approach retains the benefits of on-chip binning, including suppression of camera readout noise, faster data acquisition, smaller raw data volume, and faster processing. Use of the theoretical Raman spectral libraries eliminates the need for extensive calibrations of each matrix term, and this reduces uncertainty at conditions where calibrations are difficult. The method also allows for automatic correction of the effects of laser beam steering and optical distortion (bowing) of the laser image. The hybrid method was developed for a specific experimental setup but may be adapted to any Raman/Rayleigh scattering system that is optically well characterized.

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